

Three Ways to Judge an Agent-Native Research Artifact

A metascience comparison of GRO ideal metrics, an ARA-inferred metric design (v3), and a semantic verifier (Seal Level 2) over 12 compiled ARAs

o. Abstract

Primary result up front (§2): the core test is the substrate benchmark — the GRO extension vs the raw ARA shape with the metric held constant (H1), and the GRO metrics vs the ARA verifier (H2). The extension's value is affordance, cost, and determinism — not fidelity on metrics the ARA already encodes — and the verifier is complementary to, not beaten by, the deterministic metrics.

We compare three distinct approaches to evaluating **Agent-Native Research Artifacts (ARAs)** — machine-readable knowledge packages compiled from published papers — over a fixed corpus of **12 ARAs** in the Alzheimer's disease / neurodegeneration domain. The three approaches occupy different **rigor tiers**:

1. GRO ideal metrics (Tier A, deterministic + reliable-anchored + reproducible-judged). A battery of metrics computed over the ARA's typed graph (claims, quantities, references, entities, genre), plus a smaller set of externally anchored checks (live PubMed / registry lookups) run on the biomarker/trial half of the corpus. **2. The first ARA-inferred metric design (v3).** An earlier tournament-selected battery — `extractive_fidelity`, six `paper_rankers`, and a composite `artifact_trust` — that layers a partial source-text semantic check on top of structural completeness logic. **3. The ARA verifier (Seal Level 2).** A calibrated semantic reviewer that reads the argument and scores six epistemic dimensions (D1–D6), producing a per-ARA recommendation from Strong Accept to Reject.

Three findings organize the paper. **First**, the approaches differ enormously in *reach*: GRO scores all 12/12 ARAs; v3 fully scores only 3/12; Seal scores all 12 but through human-grade semantic labor. **Second**, they measure *near-orthogonal constructs*: Seal's overall score is essentially uncorrelated with GRO's structural health (Spearman $\rho = -0.011$) and with v3 artifact-trust ($\rho = 0.035$), while the two structural instruments correlate strongly with each other ($\rho = 0.678$, $p = 0.015$). **Third**, no single approach discriminates good science from bad science on its own — each catches real defects the other two are structurally blind to. We argue the three tiers are **complementary, not competing**: a deterministic structural floor, an externally anchored middle, and a judged semantic ceiling. We are explicit about the limits: the anchored/judged evidence exists for only half the corpus, there is **no external good-vs-bad ground-truth set**, and $n = 12$ in a single sub-domain.

1. Setup

1.1 The 12-ARA corpus

All 12 ARAs were compiled from published Alzheimer's / neurodegeneration papers and carry GRO "ideal" data shapes (a typed claim/quantity/reference/entity/genre graph). They fall into two natural clusters that matter for everything downstream:

- A **biomarker + trial cluster** (6 ARAs): che26, val25, han26, zim25, jes26, tit26 — the p-tau217 assay and donanemab-trial papers. **These are the only ARAs with any externally anchored or judged-tier evidence.**
- A **biology / epidemiology / pipeline cluster** (6 ARAs): gau25, zho25, aki26, cum26, xu25, ard25 — assessed on deterministic-tier metrics only.

Slug	Short description	Genre (GRO paper-type)	# claims
che26	Diagnostic performance of plasma p-tau217 (network meta-analysis)	systematic_review_meta_analysis	8
val25	Blood biomarkers of AD (SNAC-K observational cohort)	observational_epidemiology	13
han26	VeraBIND Tau — plasma tau pathological activity	diagnostic_accuracy_study	12
zim25	Donanemab in early symptomatic AD (long-term extension)	randomized_controlled_trial (LTE)	12
jes26	Efficacy and safety of donanemab	randomized_controlled_trial	10
tit26	Automated high-throughput plasma p-tau quantification	diagnostic_accuracy_study	11
gau25	Single-nucleus + spatial transcriptomic profiling	primary_experimental	14
zho25	Microglia networks (narrative review)	narrative_review_survey	14
aki26	Molecular signatures of resilience to AD	molecular_neuropathology_single_cell_omics	10
cum26	AD drug-development pipeline census	clinical_pipeline_registry_census	10
xu25	Epidemiological & sociodemographic transitions	epidemiological_secondary_analysis	13
ard25	Response of spatially defined microglia states	preclinical_mechanistic_study	13

1.2 The three approaches

- **GRO ideal metrics.** Eight deterministic metrics computed over the typed ARA graph, every one flagged `computable_on_prose: false` — they *only* exist because the compiler typed the graph, and they *never read the source PDF*. Plus four externally anchored/judged metrics (retractions, external reference-resolvability, prior-literature novelty, trial-registry concordance) run against live PubMed / ClinicalTrials.gov for the 6-ARA biomarker cluster.
- **v3 (first ARA-inferred design).** `extractive_fidelity` (an LLM re-reads the source and flags unsupported/contradicted/distorted claims), six `paper_rankers`, and a composite `artifact_trust` (`grounding_trust`, `environment_completeness`, `falsifiability_presence`, `compiler_fidelity`, `gate_pass_ratio`). v3 **degrades to pending rather than fabricating a score** when it lacks the input to judge.
- **Seal Level 2.** A semantic reviewer scoring six dimensions on a 1–5 scale — **D1 Evidence Relevance, D2 Falsifiability Quality, D3 Scope Calibration, D4 Argument Coherence, D5 Exploration Integrity, D6 Methodological Rigor** — then an overall score and a recommendation.

1.3 The three rigor tiers

The cleanest way to read the whole comparison is as three tiers of epistemic warrant:

Tier	What it answers	How	Primary instrument here
Tier A — deterministic	"Is the artifact internally well-formed?"	Pure lookup / arithmetic over typed fields; no comprehension, no external call	GRO deterministic metrics; v3 structural components
Tier B — reliable-anchored	"Do its external referents actually exist?"	Live query to an authority the author can't edit (PubMed, registry)	GRO retractions / external ref-resolution / registry
Tier C — reproducible-judged / semantic	"Is the underlying science any good?"	LLM/human reads the argument and judges	GRO novelty verdict (judged); Seal L2 (semantic)

The central empirical claim of this paper is that these tiers are **near-orthogonal in practice** and each is **necessary but not sufficient**.

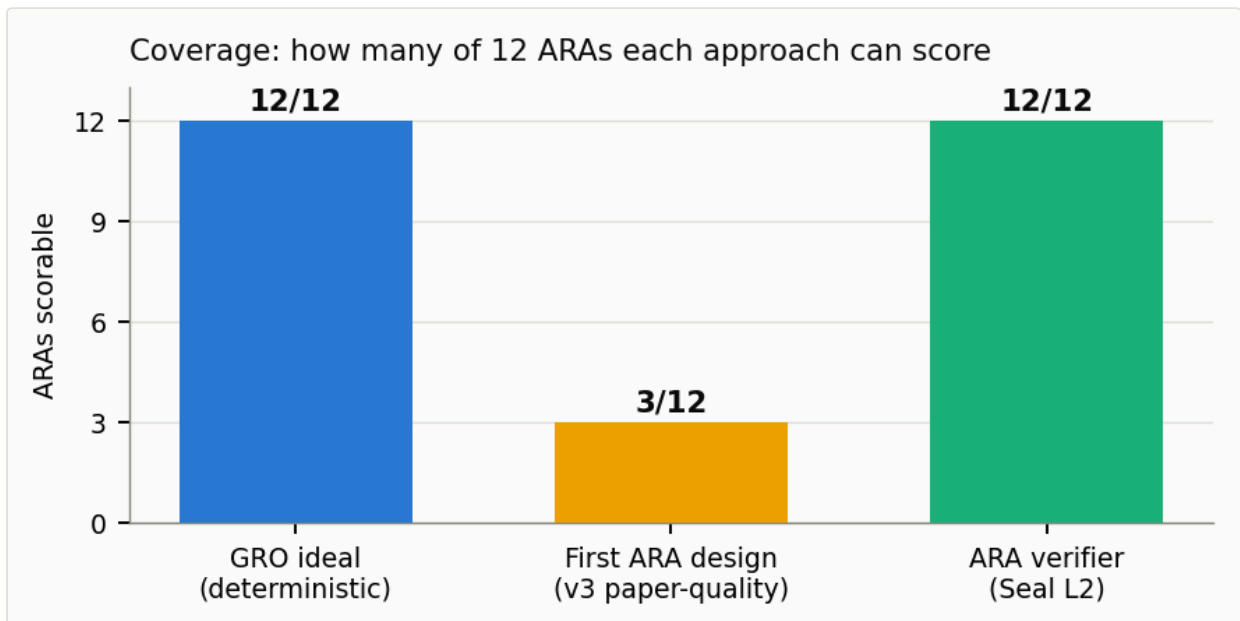


Figure 1. Coverage — how many of the 12 ARAs each approach can score.

2. Core hypothesis — the substrate benchmark (primary result)

The load-bearing test is not "GRO metrics vs v3" (that comparison, later, is secondary context — different metric *designs*). It is two sharper questions: **(H1) does the GRO extension beat the raw ARA shape when the metric is held constant?** and **(H2) how do the GRO metrics stand against the ARA verifier (Seal)?** Both were benchmarked directly (`benchmark.py`, `benchmark.json`).

2.1 H1 — GRO extension vs raw ARA, same metric, two shapes

Four of the eight GRO metrics are expressible on *both* shapes, so the identical metric logic was run over (a) the raw ARA prose (parsed from `claims.md` / `related_work.md` / `experiments.md`) and (b) the GRO typed sidecars; reference-resolvability was additionally checked against **live PubMed**.

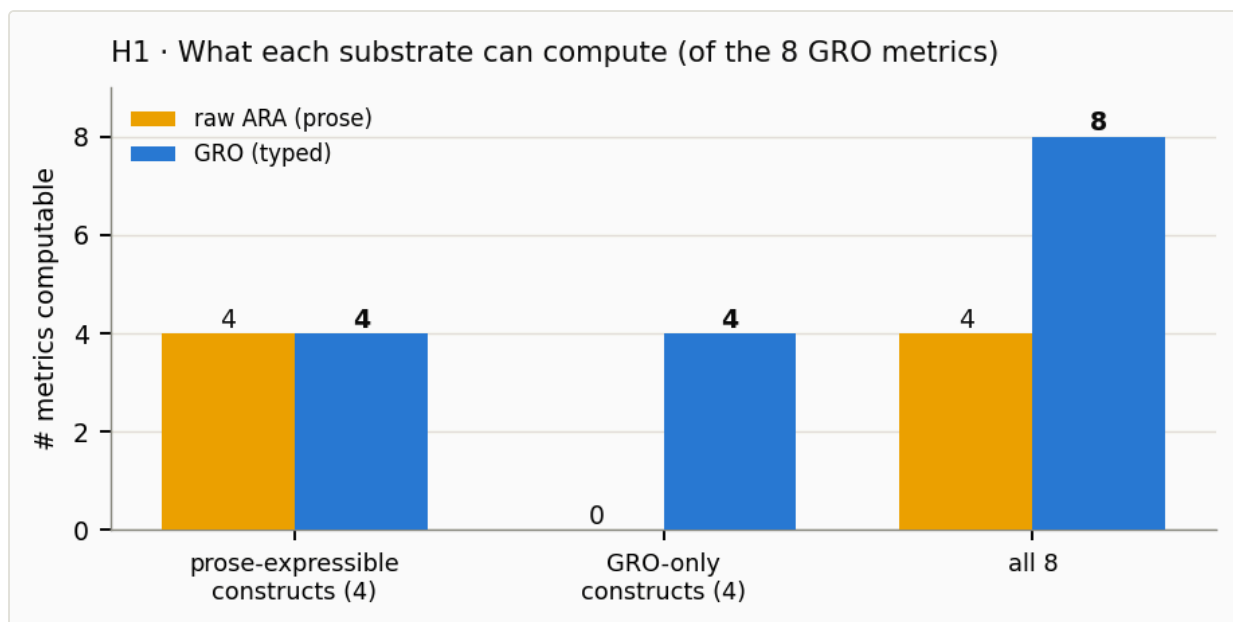


Figure 5 (H1). Of the 8 GRO metrics, the raw ARA can express 4; the typed shape enables all 8.

The extension's real win is enabling, not improving. Only 4 of the 8 GRO metrics have *any* prose form; the other 4 — claim-typing, entity-anchoring, over-claim, genre — are computable **only** on the typed shape (the raw ARA has no field to read). That is what the extension buys: metrics the substrate could not express at all.

But typing an already-expressible metric is not a free lunch — and can lose fidelity. On the 4 shared constructs the substrate *materially changes the value* (mean $|\text{prose} - \text{typed}| = 0.21 / 0.17 / 0.28$), and on the one construct with external ground truth, the **raw ARA prose is closer to PubMed truth than the typed sidecar (MAE 0.31 vs 0.44)** — the LLM extension conservatively under-declared resolvable references that the prose (carrying the actual DOIs) captured. Typing re-encodes, and re-encoding by an LLM can drift.

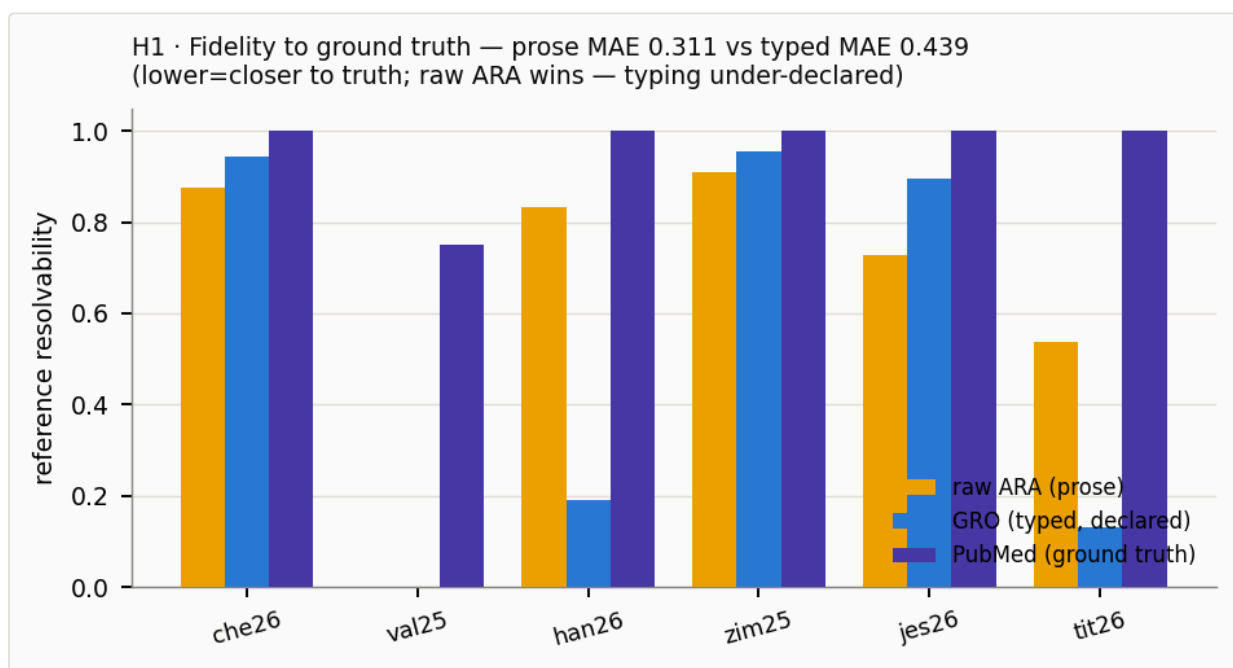


Figure 6 (H1). Reference resolvability — raw ARA prose vs GRO typed vs live-PubMed truth; prose is closer to truth.

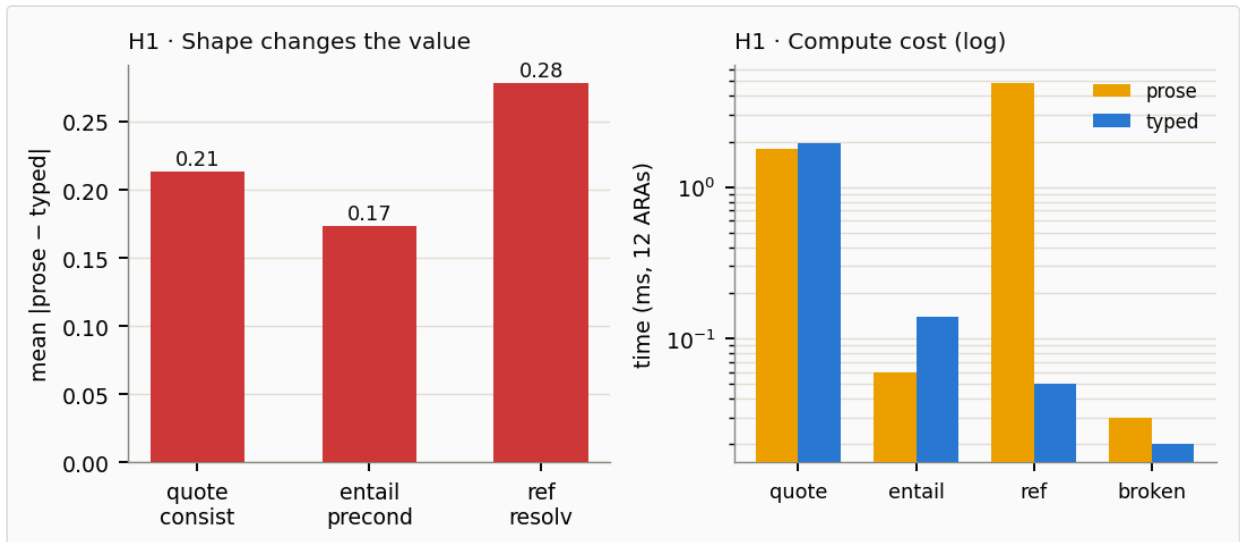


Figure 7 (H1). Left: the substrate changes the metric value (mean |prose–typed|). Right: compute cost (log ms).

Where typing wins unambiguously among the shared constructs is **cost and determinism**: reading a typed field is $\sim 100\times$ faster than parsing prose (reference resolvability 0.05 ms vs 4.9 ms across 12 ARAs) and is bit-identical on rerun.

H1 verdict. The extension earns its place by (i) making 4 previously-inexpressible metrics computable and (ii) determinism / uniformity / speed — **not** by improving metrics the raw ARA already supports, where LLM re-encoding can even reduce fidelity. "GRO > ARA" holds for *affordance and cost*, not for *fidelity on what ARA already encodes*.

2.2 H2 — ARA verifier (Seal L2) vs GRO metrics

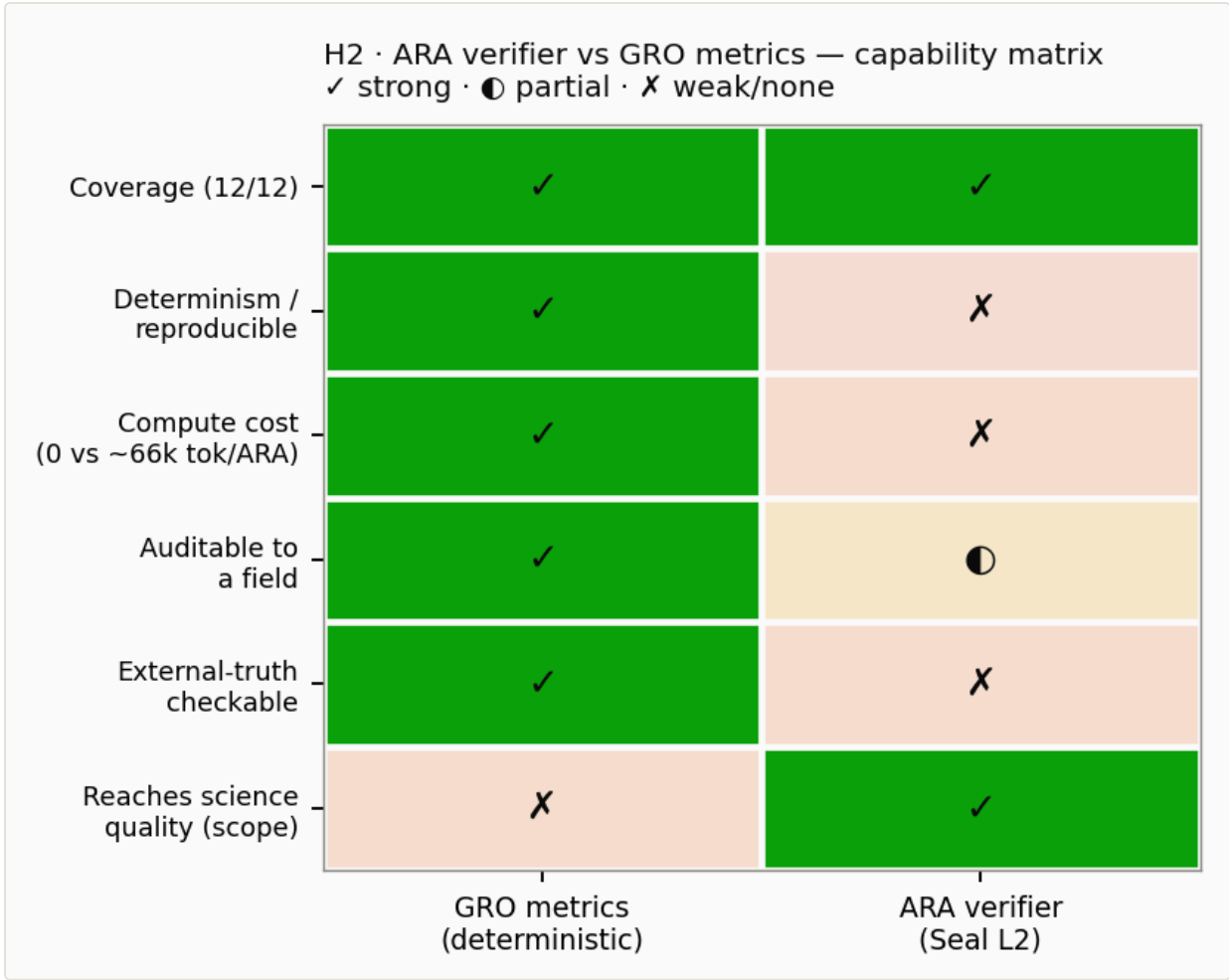


Figure 8 (H2). ARA verifier (Seal) vs GRO metrics — capability matrix.

Head-to-head on performance: the **GRO metrics win** coverage (12/12), determinism/reproducibility, compute cost (**0 LLM tokens** vs ~66k tokens per ARA for a Seal review), audibility (every score traces to a typed field), and external-truth-checkability. The **Seal wins the one dimension that most bears on science quality**: it reaches semantic judgments — scope calibration, argument coherence, evidence relevance — that the deterministic metrics are structurally blind to (it flags che26's Co5 over-generalization where GRO's over-claim count is 0).

H2 verdict. Not a winner — **complementary**. The GRO metrics are a cheap, auditable, deterministic *floor*; the verifier is the calibrated *judgement* on top. Neither, alone, separates good science from bad — and this corpus has no known-bad papers to prove either could.

3. The GRO ideal metrics: best vs. most uncertain

GRO's own analysis ranks its metrics by reliability. The top-3 are metrics that fire consistently and are hard to game; the bottom-3 are metrics whose *construct validity* is in doubt — including one that is directly falsified by GRO's own anchored check.

3.1 The three best

1. **claim_typing_coverage (deterministic)**. 1.0 across all 12/12 ARAs, 0 nulls. A pure structural presence/enum check (`claim_type ∈ enum AND logical_form ≠ not_specified`) with no external

dependency. Gaming it would require fabricating a non-empty logical form for *every* claim — a visible, checkable artifact. Zero variance across five very different genres is evidence the check is well-specified, not trivially true. **Caveat: this same zero variance means it has no discriminating power in this corpus.** 2. **quantity_reconciliation_rate (deterministic).** 1.0 across all 10/12 ARAs where it is defined (null for zho25, aki26 — structurally vacuous, no quantity cited by ≥ 2 claims). An arithmetic cross-check between two independently authored fields (the ledger's canonical value vs. the numeral embedded in the claim's own quote). The only blemish is coverage, not correctness. 3. **retractions (reliable-anchored).** 0 found across all 6 externally-checked ARAs and their top-cited references, each verified via itemized live PubMed publication-type queries. Anchored to an authority the ARA author cannot edit. **Caveat: floor effect — the corpus contains no known-bad papers, so this is proven trustworthy, not proven discriminating.**

3.2 The three most uncertain

1. **reference_resolvability_rate (self-declared / deterministic).** The one metric with a direct ground-truth check on the *same quantity* — and the ground truth contradicts it. Declared corpus mean **0.5197** vs. independently observed **0.9583** (a ≈ 0.44 gap). val25 declares 0.0 (0/55) but PubMed resolved 9/12 sampled refs; han26 declares 0.19 but resolves 1.0; tit26 declares 0.132 but resolves 1.0. Per GRO's own conclusion, the deterministic metric "scores DOI-capture completeness rather than reference verifiability" — it measures whether the compiler captured a DOI at compile time, not whether the reference is real. 2. **Prior-literature novelty verdict (reproducible-judged).** 22.6% of checked claims (7/31) land in `cannot_determine`; verdicts split almost evenly across all four categories (novel 32.3% / incremental 32.3% / cannot_determine 22.6% / previously_reported 12.9%). Each verdict rests on an LLM judgment over a PubMed search, and the methodology surfaces its own limit — "absence of a PubMed hit is not strong evidence of absence." Only 31 claims across 6/12 ARAs are covered at all. 3. **Trial-registry concordance (anchored + judged).** Only 2 trial papers exist to check (zim25, jes26), and **both are post-hoc analyses of the exact same trial, NCT04437511** — so, per the source file, "registry-concordance evidence is effectively a single data point." The concordance-gap verdict (post-hoc CDR-Global HR, plasma p-tau217, LTE-vs-ADNI comparisons added beyond registered outcomes) is real but generalizes to nothing. 10/12 ARAs have registry = null.

3.3 Also flagged (construct-validity concerns)

- **genre_silent_omissions** has a documented formula bug: `count(expected) - count(present) - count(absent_declared)` can go **negative** (zho25 = -1) when a genre's `present_slots` includes a slot outside its own `expected_slots` list.
- **broken_ref_integrity** is 100% concentrated in one ARA (ard25 = 16/16); ard25's own diagnostics attribute this to a missing `logic/experiments.md` file, not genuinely broken citations — the metric conflates "cites a nonexistent id" with "cites a real id in a ledger the compiler never found."
- **overclaim_flags** concentrates almost entirely in two genres (xu25 epidemiological projections = 11/13; cum26 pipeline census = 9/10) that structurally traffic in population-level estimates lacking a single sample-size field — raising the question of whether it flags overclaiming or just a genre feature.
- **entity_anchoring_rate** = 0.0 for all 12/12 ARAs — a corpus-wide pipeline gap (no ARA does ontology linking), reliably measured but with zero discriminating power.

3.4 Per-metric summary

GRO metric	Tier	Corpus stat	Coverage	Reliability verdict
claim_typing_coverage	deterministic	mean 1.0 (min 1.0, max 1.0)	12/12	Best — but non-discriminating
quantity_reconciliation_rate	deterministic	mean 1.0	10/12 (2 null)	Best — coverage gap only
retractions	anchored	0 found	6/12	Best — floor effect
entailment_precondition_rate	deterministic	mean 0.821 (0.4–1.0)	12/12	Middle — real spread, useful
external reference-resolvability	anchored	observed mean 0.958	6/12	Middle — sound check, small sample
genre_silent_omissions	deterministic	total 4 (min –1, max 4)	3/12 nonzero	Middle — formula defect (can go negative)
overclaim_flags	deterministic*	total 23 (max 11)	3/12 nonzero	Middle — genre-blindness / construct doubt
broken_ref_integrity	deterministic	total 16 (all ard25)	1/12 nonzero	Middle — single-ARA compilation artifact
entity_anchoring_rate	deterministic	0.0 everywhere	12/12	Middle — reliable, zero discrimination
reference_resolvability_rate	deterministic	mean 0.374 (0.0–1.0)	12/12	Most uncertain — falsified by anchored check
prior-lit novelty verdict	judged	31 claims, 22.6% cannot_determine	6/12	Most uncertain — search-phrasing dependent
trial-registry concordance	anchored + judged	2 papers, same trial	2/12	Most uncertain — n = 1 trial

*overclaim_flags is nominally deterministic but scored over LLM-typed population_scope / population_n fields.

3.5 The full deterministic grid

Every deterministic metric, every ARA. CTC and EAR are constant (1.0 and 0.0) and carry no discriminating information; the action is in the middle columns.

ARA	QuantRecon	Entailment	RefResolv	GenreOmit	Overclaim	BrokenRef
che26	1.00	1.00	0.944	0	0	0
val25	1.00	0.692	0.000	1	0	0
han26	1.00	1.00	0.190	0	0	0
zim25	1.00	1.00	0.957	0	0	0
jes26	1.00	1.00	0.895	0	0	0
tit26	1.00	1.00	0.132	0	0	0
gau25	1.00	0.571	0.100	0	0	0
zho25	null	0.500	0.000	–1	3	0
aki26	null	0.400	0.136	0	0	0
cum26	1.00	1.00	1.000	0	9	0
xu25	1.00	0.923	0.137	0	11	0
ard25	1.00	0.769	0.000	4	0	16

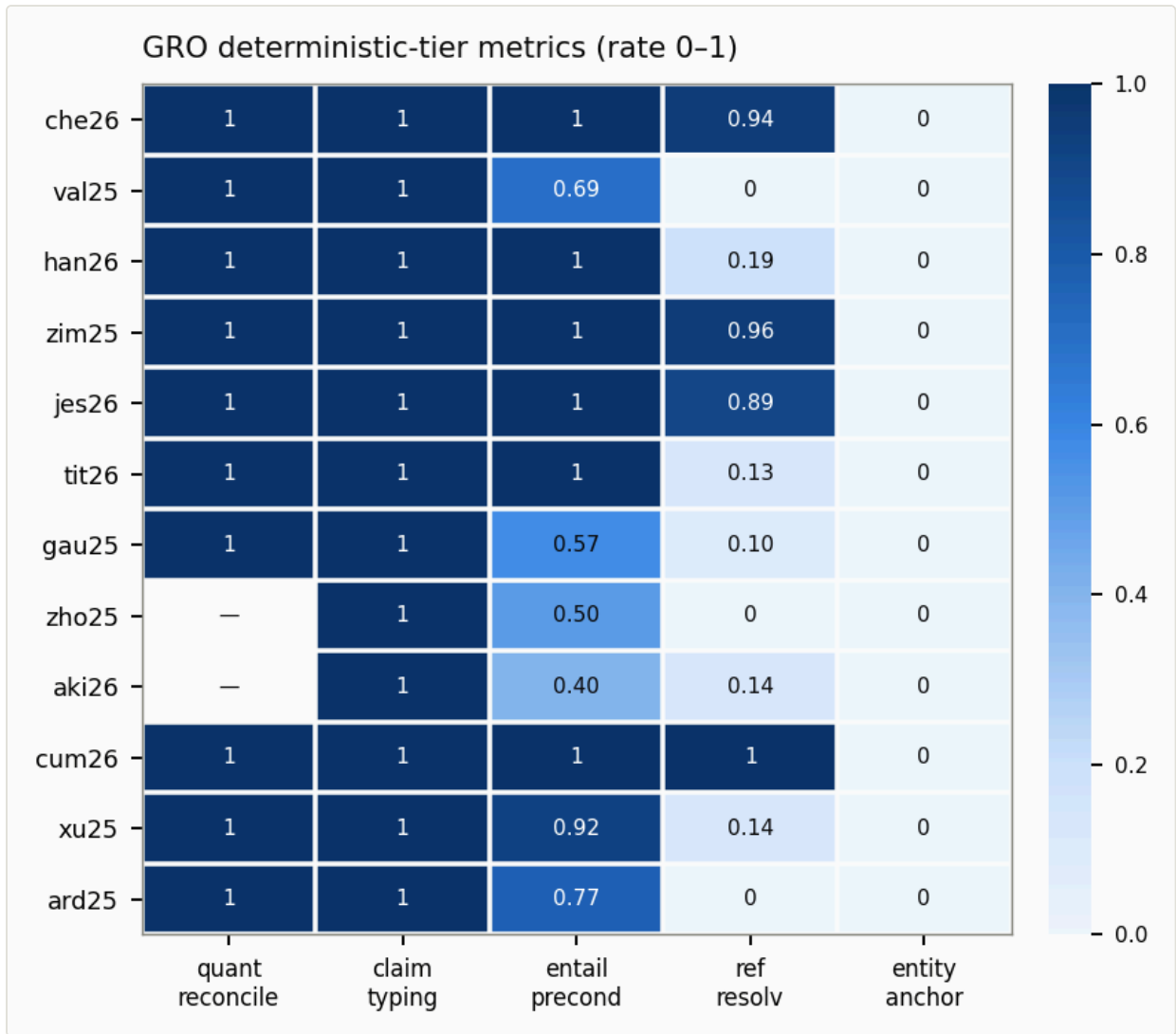


Figure 2. GRO deterministic-tier metrics per ARA (rate 0-1; “—” = undefined).

4. Held against the first ARA-inferred design (v3)

3.1 The coverage gap is architectural

v3 could **fully score 3/12 ARAs (25%)**; GRO scored **all 12/12 (100%)**.

- v3 `rank_eligible = true`: only che26, tit26, cum26.
- v3 `extractive_fidelity` computed (non-pending): 4/12 — che26 (0.625), jes26 (0.9), tit26 (1.0), cum26 (1.0); pending on the other 8/12.
- v3 `semantic_grounding` judge-scored: 4/12; pending on 8/12.
- GRO: 12/12 processed, 0 failed; the only metric-level nulls are `quantity_reconciliation_rate` on zho25 and aki26 (structurally vacuous).

The gap is not incidental. **Every GRO metric is `computable_on_prose: false` and computes over the typed graph that exists on every compiled ARA.** v3's `extractive_fidelity` and `paper_rankers` additionally require an LLM to re-read the source PDF and judge fidelity/novelty — a pass wired up (frozen judge findings) for only 3–4 ARAs. So **v3 trades reach for depth** (it verifies extraction against source prose, but only where the pass was run) while **GRO trades depth**

for reach (it runs everywhere, but only checks the ARA's own internal consistency and never sees the source paper).

3.2 Construct overlap

Construct	v3 metric	GRO metric	Relationship
Claim-to-source fidelity	<code>extractive_fidelity</code> (re-reads source PDF)	<code>quantity_reconciliation_rate</code> + <code>entailment_precondition_rate</code> (ledger self-consistency only)	GRO strictly weaker — it cannot catch a claim that is faithful to its own record but was never true of the source. v3's is the only construct in either system that re-checks the source text.
Reference / citation integrity	<code>artifact_trust.gates.all_links_resolve</code> (boolean)	<code>reference_resolvability_rate</code> (continuous) + <code>broken_ref_integrity</code> (count)	Same target; GRO more diagnostic (continuous rate/count vs. a single pass/fail bit).
Claim scope / overreach	<code>semantic_grounding</code> (judge-scored, 4/12)	<code>overclaim_flags</code> (mechanical scope-label vs. missing-n)	Different axis of the same worry; GRO's runs everywhere but is a narrow, genre-correlated proxy.
Genre-appropriate completeness	genre used only as an eligibility label	<code>genre_silent_omissions</code>	GRO adds this construct outright; v3 has no analog.

GRO-only constructs (no v3 analog): `entity_anchoring_rate` (0.0 corpus-wide), `broken_ref_integrity`, `genre_silent_omissions`, `overclaim_flags`. **v3-only constructs (no GRO analog):** `corrective_science_score`, `dead_end_density` / `negative_result_share`, `translation_trial_linkage`, `novel_claim_count` — the paper rankers.

v3's own tournament headline — **0/6 usable paper rankers** — is confirmed on this corpus: `corrective_science_score` is `invalid_fabricated` on 12/12; `negative_result_share` is `pending_sem` on 12/12; `translation_trial_linkage` is self-annotated "superseded" on 12/12; `dead_end_density` is `fabricated` or `vacuous`; `semantic_grounding` is judge-scored on only 4/12; `novel_claim_count` is `pending_sem` even where a value exists. None reaches a fully valid corpus-wide state.

3.3 Agreement and divergence

- **Sharpest divergence — che26.** v3 `extractive_fidelity` = 0.625 catches a **fabricated forest-plot statistic** (MD = 0.10, 95% CI 0.04–0.16) that appears nowhere in the source. GRO scores this same ARA `quantity_reconciliation_rate` = 1.0 and `entailment_precondition_rate` = 1.0 — clean — because those metrics only check the ledger against its own quote and never touch the PDF. **GRO's structural checks cannot substitute for source verification even where both nominally target "fidelity."**
- **Second-sharpest divergence — cum26.** v3's *best* ARA (fidelity 1.0, `semantic_grounding` 0.941, 8/8 NCTs verified) is GRO's *most-overclaiming* ARA (`overclaim_flags` = 9/10). Perfectly faithful extraction of a source that itself makes unscoped population claims.
- **tit26.** v3 marks it clean (fidelity 1.0, all gates pass), but GRO's continuous `reference_resolvability_rate` = 0.132 exposes that most references are unverifiable — a distinction v3's boolean gate cannot make.

- **Sharpest agreement — ard25.** v3 gates (`seal_L1 = false`, `all_links_resolve = false`, `low_trust = true`) and GRO counts (`broken_ref_integrity = 16`, `genre_silent_omissions = 4`, both corpus maxima) converge on the same "structurally worst" verdict — GRO supplying the auditable counts v3's booleans compress away.

5. Held against the ARA verifier (Seal L2)

4.1 Per-ARA Seal scorecard

Seal dimensions: **D1** Evidence Relevance · **D2** Falsifiability Quality · **D3** Scope Calibration · **D4** Argument Coherence · **D5** Exploration Integrity · **D6** Methodological Rigor (each 1–5).

ARA	D1	D2	D3	D4	D5	D6	Overall	Recommendation	GRO health	v3 trust
che26	4	4	3	4	4	3	3.7	Weak Accept	0.982	0.944
val25	5	4	5	5	4	5	4.7	Strong Accept	0.564	0.615
han26	4	4	4	4	5	3	4.0	Weak Accept	0.730	1.000
zim25	4	3	4	5	4	2	3.7	Weak Accept	0.986	0.931
jes26	4	4	4	5	3	3	3.8	Weak Accept	0.965	0.974
tit26	4	4	4	5	3	4	4.0	Accept	0.711	0.917
gau25	4	4	5	5	4	4	4.3	Accept	0.557	0.685
zho25	4	3	4	4	2	2	3.2	Weak Accept	0.143	0.178
aki26	4	4	4	5	5	4	4.3	Accept	0.268	0.919
cum26	4	3	4	5	4	4	4.0	Accept	0.550	0.970
xu25	4	4	4	5	5	3	4.2	Accept	0.264	0.754
ard25	4	3	4	4	1	2	3.0	Reject	0.000	0.150

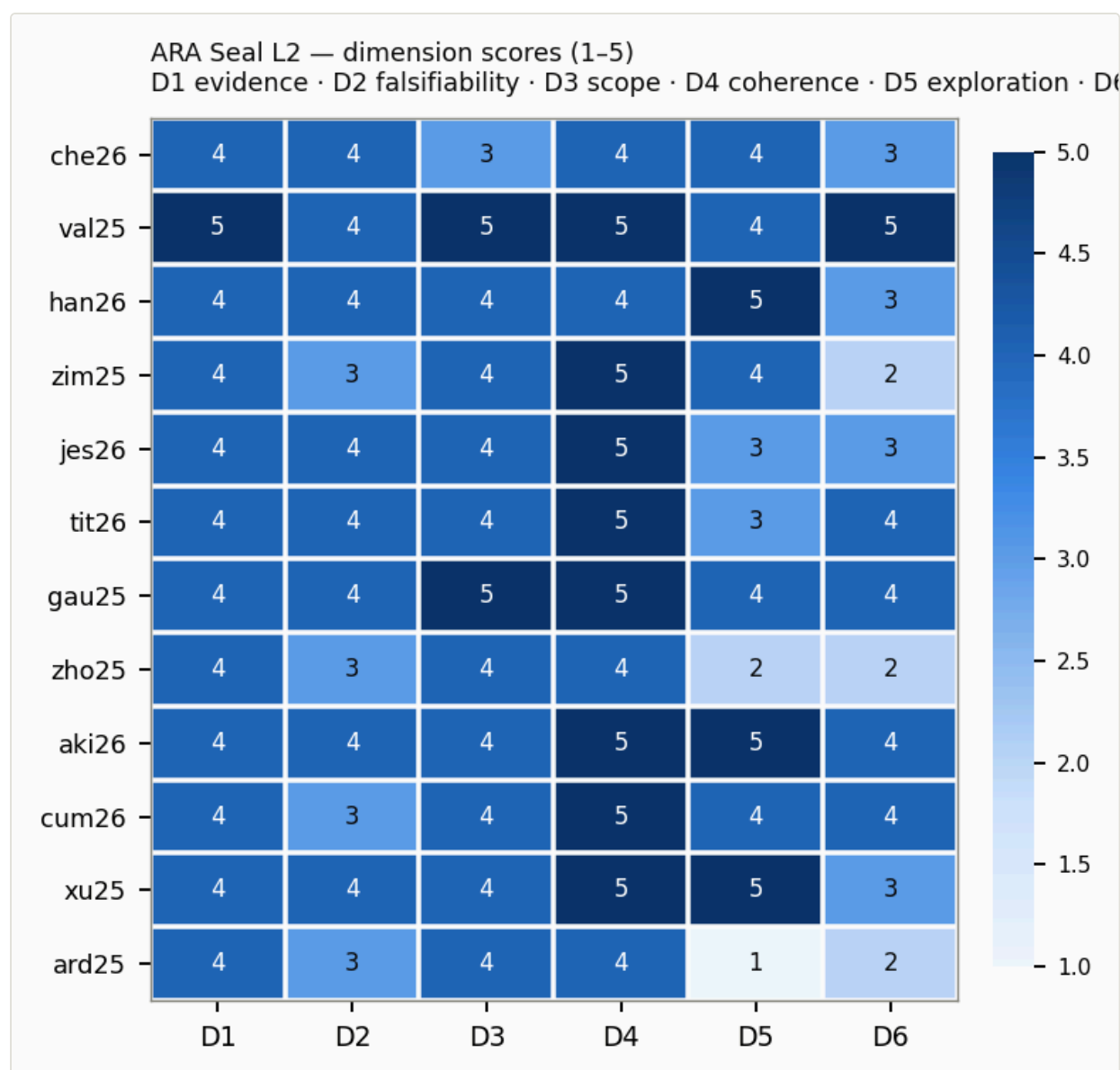


Figure 3. ARA Seal L2 dimension scores (1–5) per ARA.

4.2 What Seal catches that neither metric set can

- ****Scope over-generalization with a *stated* n (che26, D3 = 3).**** GRO `overclaim_flags` = 0 here, because its rule fires only when `population_n` is literally missing. Seal flags C05 verbatim: "*a generalization claim that overreaches its evidence type ... a P-score of 1.00 computed within only two Han Chinese cohorts (Huashan N=297, Greater Bay N=425) with no reported formal ethnicity-by-biomarker interaction test.*" The cohort sizes are named, so the structural rule is satisfied — detecting the overreach requires knowing what statistical test the claim needs. Comprehension, not a field lookup.
- **Evidence-source mismatch (che26).** Co3's "reduce the need for tau-PET scans by ~80%" is "*sourced from an unrelated external reference (Therriault et al., 2021) ... not derived from this NMA's own AUC/P-score outputs.*" No structural field encodes argument coherence.
- **Uncorrected multiplicity (jes26, zim25, val25).** A recurring methodological judgment: "*No multiplicity correction is applied ... yet every claim is nonetheless labeled 'supported' with uniform confidence.*" No GRO or v3 field maps to it.

- **Sponsor / conflict-of-interest and reproducibility (zim25, D6 = 2).** *"Sponsor-only authorship (all 18 authors are Eli Lilly employees) ... no released analysis code, no stated software version ... several supplementary figures/tables ... absent from the provided PDF."* No structural field maps to authorship or funding conflicts.

4.3 What the deterministic metrics catch that Seal misses or softens

- **Citation-graph integrity — no Seal dimension asks it.** val25 has **0/55 references resolvable** (reference_resolvability_rate = 0.0), yet Seal's D1 = 5 and overall = 4.7 (Strong Accept). Not a contradiction — D1 asks whether quoted evidence matches the claim, not whether the works-cited ledger resolves externally — but a reader of the Seal alone would never learn the bibliography doesn't resolve. Same blind spot: zho25 (0.0), ard25 (0.0), gau25 (0.1), tit26 (0.132).
- **Exact auditable counts where Seal narrates.** ard25: GRO broken_ref_integrity = 16/16 (100% of proof_refs). Seal converges narratively — *"every claim's verification-plan linkage is a dangling reference"* (D5 = 1) — but GRO supplies the precise fraction Seal's prose compresses into a floor-rule Reject.
- **Genre-blindness cutting the other way.** GRO overclaim_flags fires on 9/10 (cum26) and 11/13 (xu25) claims; Seal's D3 = 4 for both does **not** corroborate an overclaiming problem — these are census/GBD genres that legitimately state population-level estimates without a sample size. Here the *deterministic* metric is the one over-flagging.
- **A formula breaking at a genre edge.** zho25 genre_silent_omissions = -1 (a nonsensical negative from the literal count formula on a review genre); Seal instead reasons in prose that exploration integrity is *"honestly disclaimed rather than fabricated ... effectively unscorable"* (D5 = 2). Genre-aware judgment the rigid formula cannot make.

4.4 Do the three rank consistently?

Spearman rank correlation across the 12 ARAs:

Pair	ρ (all 12)	ρ	ρ (excl. ard25, n=11)
Seal overall vs GRO health	-0.011	0.974	-0.318
Seal overall vs v3 artifact-trust	0.035	0.913	-0.258
GRO health vs v3 artifact-trust	0.678	0.015	0.582 (p = 0.060)

Seal is **essentially uncorrelated** with both structural instruments, which correlate strongly with **each other**. The single point of three-way agreement is **ard25** — a structurally broken ARA (empty trace, missing logic/experiments.md) that every instrument independently ranks last. Remove it, and the Seal-vs-structural correlations trend **negative**, while GRO and v3 stay positive with each other. The two mechanical instruments agree about what "complete" looks like; neither agrees with what the calibrated reviewer considers "good."

Illustrative inversions:

- **val25** — Seal's #1 (4.7, Strong Accept) but #6 on GRO health and #10 on v3 trust (0% ref resolvability, environment_completeness 0.143). Faithfully scoped, well-evidenced, rigorously falsifiable — with real structural holes.
- **zim25** and **che26** — #1–2 on both structural instruments (GRO 0.986 / 0.982) but only #9–10 on Seal, capped by D6 rigor concerns (sponsor-only authorship, uncorrected multiplicity)

invisible to structural metrics.

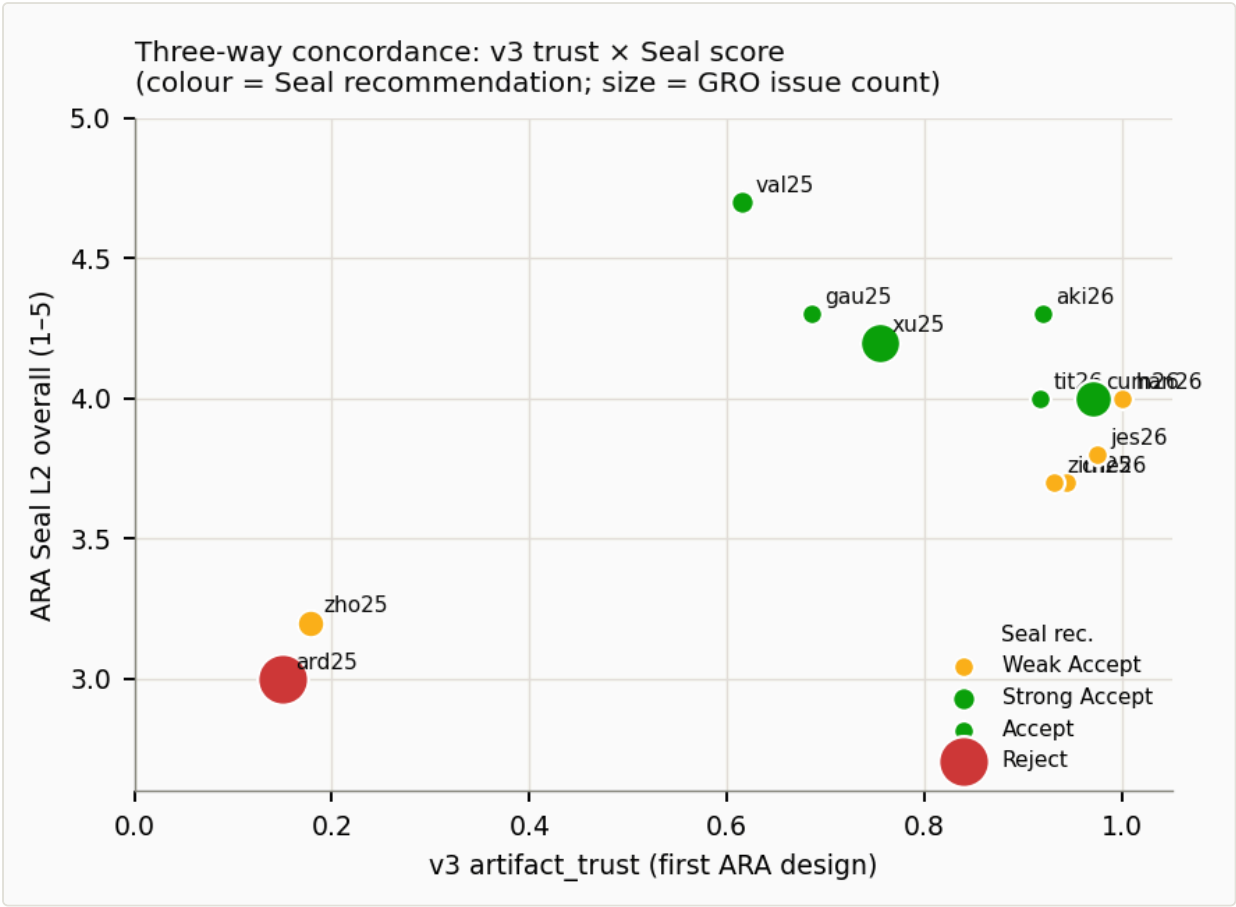


Figure 4. Three-way concordance: v3 artifact_trust × Seal overall; colour = Seal recommendation, size = GRO issue count.

4.5 The sharpest catch-divergences

Case	v3 (first ARA design)	GRO ideal metric	ARA Seal L2	What it exposes
che26 forest- plot stat	extractive_fidelity 0.625 — catches a fabricated statistic (reads source)	quantity_reconciliation 1.00 — blind (ledger-internal)	D6 rigor = 3	determinism is ledger-internal; source fabrication needs the anchored/judged tier
cum26	extractive_fidelity 1.00 (v3's best)	overclaim_flags 9/10 (GRO's worst)	4.0 Accept	faithful extraction ≠ calibrated scope
xu25	artifact_trust 0.75	overclaim_flags 11/13	4.2 Accept	mechanical over-claim flag fires on a genre feature Seal judges benign
che26 Co5	—	overclaim_flags 0 (misses it)	D3 scope 3 — flags over-generalization	Seal catches scope miscalibration the deterministic flag cannot

ARA	Instrument that catches it	What it catches	The other two
che26	v3 only	Fabricated forest-plot statistic (MD 0.10, CI 0.04–0.16)	GRO 1.0/1.0 (blind); Seal flags a <i>different</i> issue (Co5 scope)
che26	Seal only (D3)	Cross-ethnic overgeneralization from 2 cohorts, no interaction test	GRO overclaim = 0 (n is stated); v3 silent
val25	GRO only	0/55 references resolvable	Seal D1 = 5, overall 4.7; v3 pending
cum26/xu25	GRO flags, arguably wrongly	9/10 and 11/13 "overclaims"	Seal D3 = 4 (genre-appropriate); v3 clean
zim25	Seal only (D6)	Sponsor-only authorship, no code, missing supplements	GRO 0.986 (#1); v3 0.931
ard25	All three	Missing experiments ledger → dangling proof refs	Three-way agreement (the lone case)

6. Synthesis: three complementary tiers

The correlation structure is not a defect in any instrument — it is the empirical signature of the tier gap.

- **Tier A (deterministic — GRO structural, v3 structural components)** answers pure lookups over typed fields: does this ref resolve, does this claim carry a logical form, does this quantity match its quote. Zero comprehension of scientific quality. It catches **catastrophic breakage** (ard25) that no reviewer should read past, and it surfaces **facts a semantic rubric never asks about** (broken-ref counts, 0%-resolvable bibliographies). It is **necessary** — but it correlates near-zero with quality.
- **Tier B (reliable-anchored — GRO retractions / external ref-resolution / registry)** answers "do the external referents exist," via authorities the author cannot edit. It is the tier that **falsified GRO's own deterministic reference_resolvability_rate** (declared 0.52 vs. observed 0.96). Its power in *this* corpus is capped by a floor effect: no known-bad papers, so it is proven trustworthy, not proven discriminating.
- **Tier C (judged / semantic — GRO novelty verdict, Seal L2)** reads the argument and asks whether claims overreach their evidence and whether the methodology is rigorous for the genre. It catches scope overreach, multiplicity risk, and conflict-of-interest that **no structural field encodes**. It is the only tier that tracks "good science" — and it is the most expensive and the least reproducible.

No single approach discriminates good-vs-bad alone. v3's `extractive_fidelity` caught a fabricated statistic that GRO structurally cannot see (che26). GRO caught a 0%-resolvable bibliography that Seal's rubric never examines (val25). Seal caught a cross-ethnic overgeneralization and a sponsor-COI that neither metric can represent (che26, zim25). The three are near-orthogonal signals, and an ARA pipeline needs **Tier A/B structural gates before Tier C semantic review** — a deterministic floor to reject the structurally broken cheaply, an anchored middle to confirm external referents, and a judged ceiling for the epistemic call — **not one instead of the others**.

7. Limitations

We are deliberately conservative about what this comparison establishes.

1. **No external good-vs-bad ground-truth set.** The corpus contains **zero known-bad papers** — 0 retractions, no concealed outcome-switching, all references ultimately real. High reliability on retractions and external ref-resolution reflects a **floor/ceiling effect on a uniformly legitimate corpus**, not proof any instrument would correctly flag a fabricated or retracted paper. GRO's external validation says so directly: this "validates external computability and surfaces concordance/compilation gaps, but it is NOT a full good-vs-bad discrimination test." 2. **The anchored/judged tiers exist for only half the corpus.** All Tier-B/Tier-C metric evidence comes from the 6-ARA p-tau217/donanemab cluster. Every "most uncertain" verdict about anchored/judged metrics is itself demonstrated on **half the corpus, in one narrow clinical/biomarker sub-domain**. The trial-registry concordance finding rests on a **single trial** (NCT04437511) analyzed by two papers. 3. **LLM-authored extensions carry their own uncertainty.** Seal L2 is an LLM semantic reviewer; GRO's novelty verdicts are LLM judgments over PubMed searches whose outcomes are **phrasing-dependent** (22.6% cannot_determine); overclaim_flags is scored over LLM-typed fields. None of these has an independent human adjudicator in this study, so the Tier-C signal is itself unvalidated against expert consensus. 4. **Small, single-domain n.** Twelve ARAs, all Alzheimer's / neurodegeneration, split 6/6 across two clusters. The Spearman correlations ($\rho = -0.011, 0.035, 0.678$) are estimated on $n = 12$ ($n = 11$ excluding ard25); confidence intervals are wide and the near-zero Seal-vs-structural result should be read as "no detectable positive relationship in this sample," not "proven independence." 5. **Known metric defects remain unfixed in the reported numbers.** genre_silent_omissions can go negative (zh025 = -1); broken_ref_integrity conflates missing-ledger with broken-citation (ard25); the deterministic reference_resolvability_rate measures DOI-capture, not verifiability. These are reported as-is for honesty, not corrected.

The defensible conclusion is narrow and, we think, robust: **structural completeness is necessary but not sufficient, the three tiers measure near-orthogonal things on this corpus, and a credible ARA-evaluation pipeline must run all three rather than treat any one as a proxy for the others.**